

“We should stop training radiologists now. Its just completely obvious that within five years, deep learning is going to do better than radiologists.”

– Geoffrey Hinton (in 2016)

1 Introduction

Artificial intelligence (AI) is a general-purpose technology with transformative potential similar to that of the steam engine and electricity (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2017; Agrawal et al., 2018; Acemoglu and Johnson, 2023; Goldfarb et al., 2023; Frank et al., 2019). But, in contrast to the transformation during the industrial revolutions, AI can potentially displace humans from tasks that require complex reasoning (Webb, 2019; Felten et al., 2019; Brynjolfsson and Mitchell, 2017). Advances in AI have therefore caused concern about the role of human work, even in highly skilled occupations (Ford, 2015). Indeed, a growing literature shows that AI tools based on machine learning can outperform humans on a host of predictive tasks, including those typically performed by experts (Liu et al. 2019; Lai et al. 2021; Mullainathan and Obermeyer 2019; Kleinberg et al. 2017).

However, Hinton’s prediction about radiology — an iconic example in machine learning — may have been premature.¹ Many argue that instead of being replaced by AI, it is optimal to have human radiologists use AI assistance, at least for the foreseeable future (Langlotz, 2019; Agrawal et al., 2019). In addition to considerable legal and regulatory challenges that stand in the way of full automation, there may be potential gains from combining human expertise with AI input that cannot be realized by exclusively relying on one or the other. For example, radiologists may correct mistaken AI predictions or may have access to information about the clinical context that has not yet been systematized and built into AI tools. Current regulatory practice is consistent with these arguments – approved AI tools for clinical decision-making typically play a supporting role rather than operating autonomously (see Norden and Shah, 2022; Harvey and Gowda, 2020, for example). Similar reasons are likely to result in partial task automation or collaboration between human experts and AI being optimal in other contexts.

We run an experiment to examine how radiologists use AI predictions, whether AI assistance improves their performance, estimate a model of (potentially biased) belief updating, and analyze what this model implies about the optimal form of collaboration between AI and humans.² If humans correctly combine AI predictions with their own information, then AI

¹A more nuanced but qualitatively similar prediction that machine learning tools will displace radiologists is conveyed in Obermeyer and Emanuel (2016).

²We will use the terms ‘humans’, ‘radiologists’, and ‘participants’ interchangeably.

assistance unambiguously improves prediction and decision quality. Unless the costs in terms of human effort outweigh these benefits, it is optimal to provide AI assistance to humans. However, substantial literature in economics suggests that humans may err when making probabilistic judgments by deviating from the benchmark model of Bayesian updating with correct beliefs (see [Benjamin et al., 2019](#), for a review). The optimal approach for combining human and AI information in the presence of these mistakes is non-trivial.

The experiment employs professional radiologists whom we recruit through teleradiology companies to diagnose retrospective patient cases. We experimentally manipulate the information set radiologists have access to when making decisions in a two-by-two factorial design. In the minimal information environment, we provide only the chest X-ray image to which we add either AI predictions or contextual information, or both. The AI information treatment provides probabilities that a patient case is positive for a potential chest pathology generated using an algorithm trained on over 250,000 X-rays with corresponding disease labels ([Irvin et al., 2019](#)). This algorithm was shown to perform comparably to board-certified radiologists. The contextual information treatment provides clinical history information that radiologists typically have available but, for privacy reasons, was not available to train the AI. This information includes the treating doctors’ indications, the patient’s vitals, and the patient’s labs.

We first estimate the treatment effects of those informational interventions on radiologists’ prediction accuracy and the probability of making a correct decision. We then test alternative models of biased belief updating to investigate whether radiologists exhibit systematic deviations from a Bayesian benchmark when incorporating AI predictions. For example, humans may suffer from automation bias, a tendency to place more weight on machine-provided predictions than on one’s own information.³ Additionally, humans may treat AI predictions as independent of their own information ([Enke and Zimmermann, 2019](#)). Finally, we evaluate various forms of human-AI collaboration – in terms of diagnostic performance and costs of expert time – that decide, as a function of the AI prediction, to use only the AI input, only the human input, or the human input with access to AI.

Diagnostic radiology is an ideal laboratory for investigating human-AI interaction for various reasons. Deep learning has made significant advances in radiology, surpassing human decision-making in many cases, and such algorithms are already clinically deployed [Rajpurkar and Lungren \(2023\)](#). It, therefore, serves as a useful leading indicator for other professions

³This terminology is borrowed from the literature that dates to the proliferation of computerized automated support systems in aviation, which raised concerns about human complacency or automation bias (see [Alberdi et al., 2009](#), for an overview).

where similar developments will follow. Moreover, radiology is a highly-paid medical specialty, which means that the potential benefits from productivity enhancements through AI are large. Radiology is also ideal from a research perspective. Unlike other physicians, radiologists do not have direct contact with patients, allowing us to run a decision experiment that resembles their normal workflow through a remote interface that we developed. To aid quantitative analysis, instead of obtaining a free-text report, we collect radiologist’s assessed probability that a given chest pathology is present and a binary clinical recommendation of whether to treat or follow up for that pathology. For our experiment, we hired 180 radiologists through teleradiology companies that serve US hospitals and offer the services of both US-based and non-US-based radiologists.

Analyzing the quality of our participants’ assessments requires establishing, for each patient case, a standard or ground truth. An important challenge in our setting is that definitive diagnostic tests do not exist for most thoracic pathologies and, even when they do exist, are selectively performed depending on a radiologist’s recommendation. We therefore follow the machine learning literature (Sheng et al., 2008) and construct a diagnostic standard by aggregating the assessments of five board-certified radiologists at Mt. Sinai Hospital with at least ten years of experience and chest radiology as a sub-specialty. We assess the robustness of all our results by constructing a (leave-one-out) ground truth using the assessments of our experimental participants and by varying the aggregation method.

There are two important challenges in designing the experiment, which we address using a combination of experimental designs. First, while teleradiology firms allow us to recruit radiologists for our experiment we need to compensate them at the market rate, making data collection expensive. An across-participant design would therefore be cost-prohibitive except for extremely large effect sizes. In the first design, we, therefore, randomize each participant into an order in which they are exposed to the four informational environments but do not encounter cases repeatedly. Thus, this design allows for within-participant comparisons in addition to an initial pure across-participant comparison. The second challenge is that, to estimate a model of belief updating, we need to observe radiologists’ assessments of a given case both with and without AI assistance. Moreover, the assessment without AI assistance is required to estimate the Bayesian benchmark, which we can compare to the assessment that incorporates AI assistance. Our second design addresses this issue by asking each participant to diagnose each patient case once in each of the four information environments in random order while ensuring at least a two-week wash-out period between two encounters of a given case. This wash-out period is intended to eliminate the memory of prior information and